Dynamic Programming – Basic Problems

Dice Throw Problem, Binomial Coefficient, Assembly Line Scheduling

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Dice Throw Problem

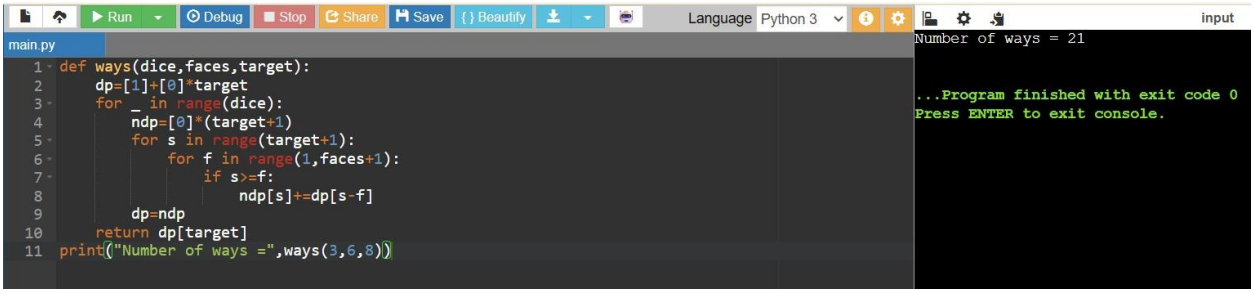
1. Write a program to determine the number of ways a player can unlock a reward in an online game. The game uses 3 six-faced dice, and the reward is granted only when the total score obtained is exactly 8. Display the total number of possible combinations that produce the target score.

Input:

Number of Dice = 3
Faces per Dice = 6
Target Score = 8

Output:

Number of ways = 21



Question 2: Board Game Tournament

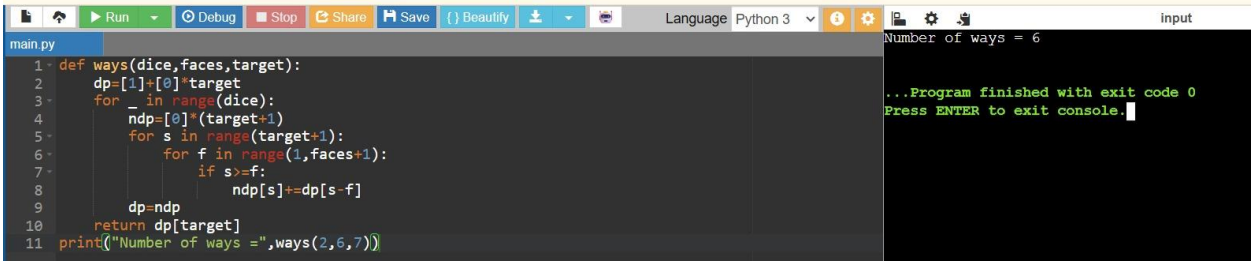
Write a program to calculate the number of possible outcomes in a board game tournament where a player rolls 2 dice, each having 6 faces. To qualify for the next round, the player must obtain a total score of 7.

Input:

Number of Dice = 2
Faces per Dice = 6
Target Score = 7

Output:

Number of ways = 6



Question 3: Warehouse Robot Navigation

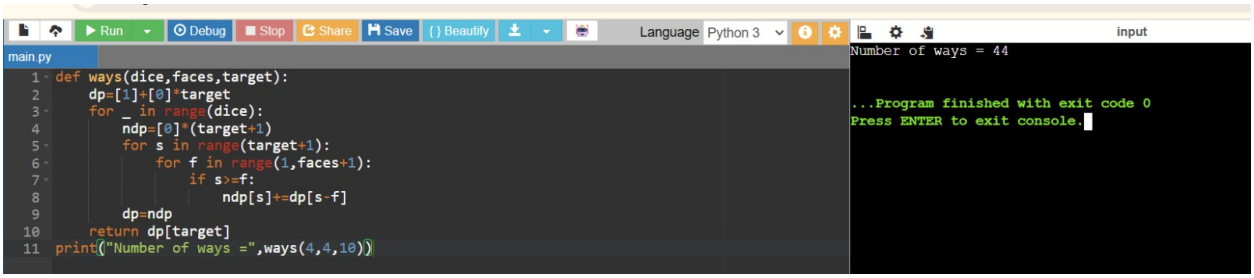
A warehouse robot moves forward based on the values obtained from 4 dice throws. Each dice has 4 faces. The robot reaches its destination only if the sum of all throws equals 10. Write a program to determine the number of possible movement sequences.

Input:

Number of Dice = 4
Faces per Dice = 4
Target Distance = 10

Output:

Number of ways = 44



Question 4: Casino Reward Calculation

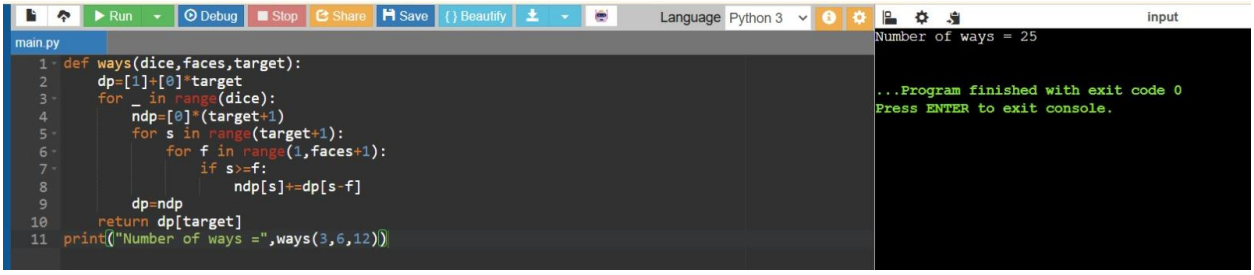
A casino rewards customers who obtain exactly 12 points by rolling 3 six-faced dice. Write a program to calculate the total number of winning combinations.

Input:

Number of Dice = 3
Faces per Dice = 6
Target Sum = 12

Output:

Number of ways = 25



Question 5: Quiz Competition Scoring

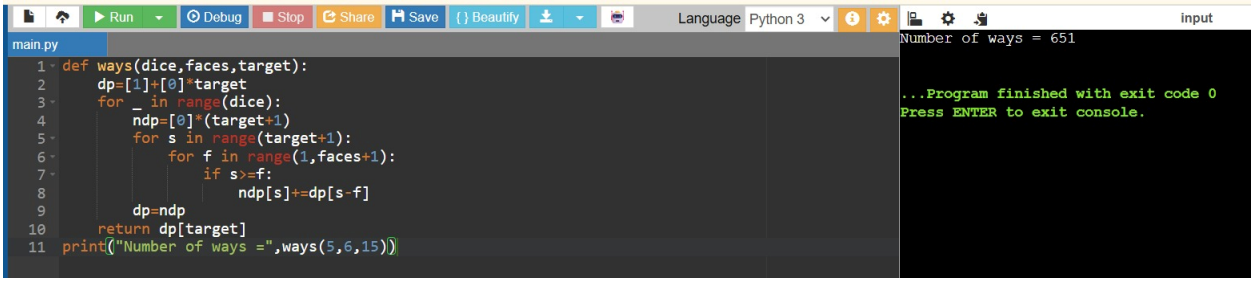
In a quiz application, bonus points are awarded if the sum of 5 dice rolls equals 15. Each die has 6 faces. Write a program to find the number of valid scoring combinations.

Input:

Number of Dice = 5
Faces per Dice = 6
Target Score = 15

Output:

Number of ways = 651



Question 6: Sports Prediction System

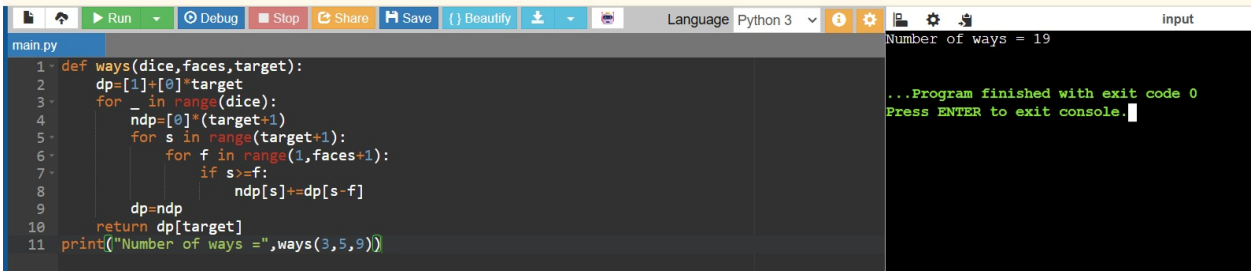
Write a program to determine the number of ways a predicted score of 9 can be obtained using 3 five-faced dice.

Input:

Number of Dice = 3
Faces per Dice = 5
Target Score = 9

Output:

Number of ways = 19



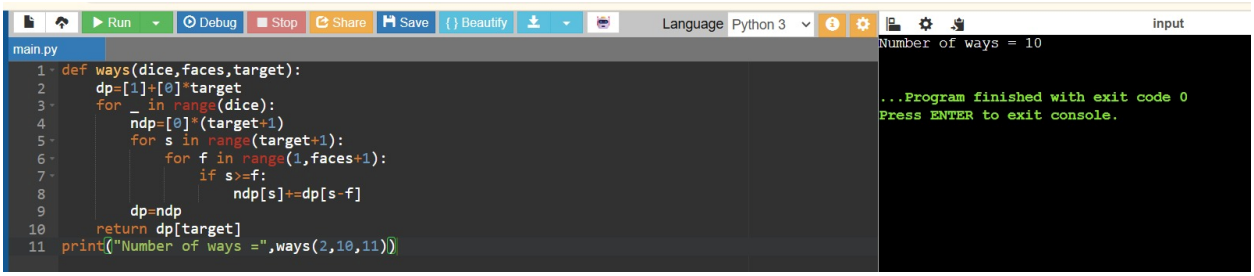
7. A lottery simulation system uses 2 ten-faced dice. Find the number of possible combinations that result in a sum of 11.

Input:

Number of Dice = 2
Faces per Dice = 10
Target Sum = 11

Output:

Number of ways = 10



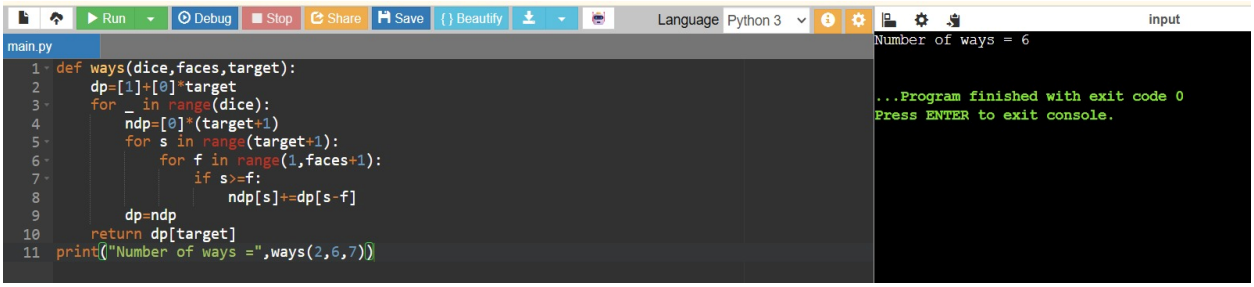
8.Write a program to find the number of ways to obtain a given sum using N dice, each having M faces.

Input:

Number of Dice = 2
Faces per Dice = 6
Target Sum = 7

Output:

Number of ways = 6



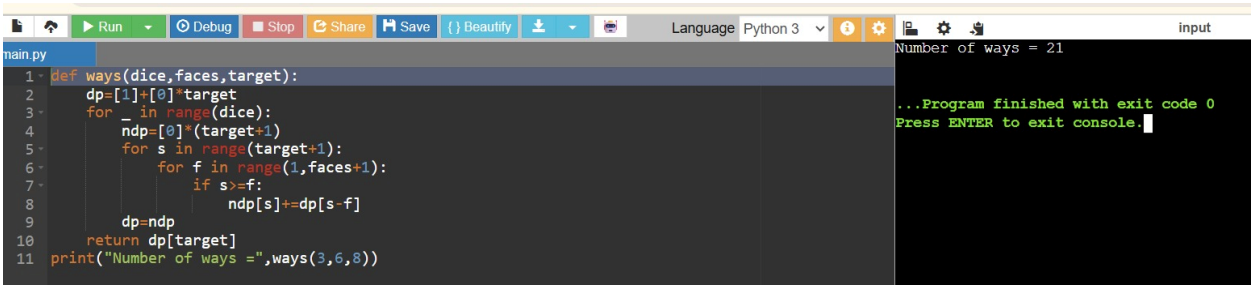
9.Write a program to determine the number of ways to obtain a target score using multiple dice.

Input:

Number of Dice = 3
Faces per Dice = 6
Target Sum = 8

Output:

Number of ways = 21



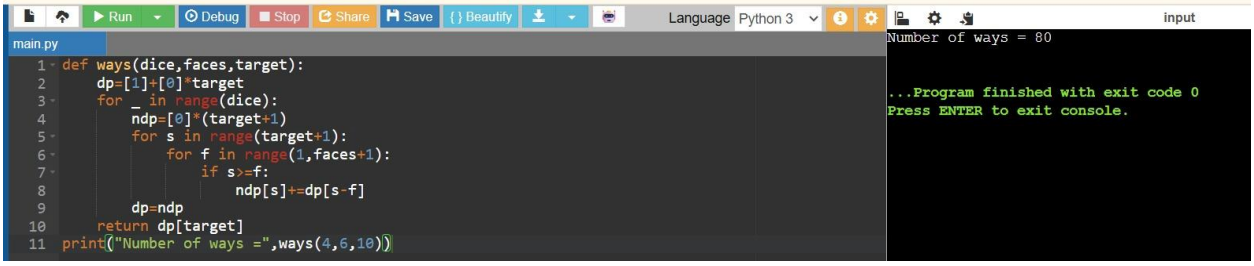
10.Write a program to count the possible outcomes that produce a specified sum.

Input:

Number of Dice = 4
Faces per Dice = 6
Target Sum = 10

Output:

Number of ways = 80



11.Write a program to find the number of possible dice combinations that yield the required score.

Input:

Number of Dice = 2
Faces per Dice = 4
Target Sum = 5

Output:

Number of ways = 4

```
def ways(dice,faces,target):
    dp=[1]+[0]*target
    for _ in range(dice):
        ndp=[0]*(target+1)
        for s in range(target+1):
            for f in range(1,faces+1):
                if s>=f:
                    ndp[s]+=dp[s-f]
        dp=ndp
    return dp[target]
print("Number of ways =",ways(2,4,5))
```

```
Number of ways = 4

...Program finished with exit code 0
Press ENTER to exit console.
```

Binomial Coefficient

1.Write a program to determine the number of ways to obtain the target sum using dice throws.

Input:

Number of Dice = 3
Faces per Dice = 4
Target Sum = 7

Output:

Number of ways = 12

```
def ways(dice,faces,target):
    dp=[1]+[0]*target
    for _ in range(dice):
        ndp=[0]*(target+1)
        for s in range(target+1):
            for f in range(1,faces+1):
                if s>=f:
                    ndp[s]+=dp[s-f]
        dp=ndp
    return dp[target]
print("Number of ways =",ways(3,4,7))
```

```
Number of ways = 12

...Program finished with exit code 0
Press ENTER to exit console.
```

2.Write a program to calculate the Binomial Coefficient C(n,r).

Input:
n = 5
r = 2
Output:
C(5,2) = 10

```
def combination(n,r):
    r=min(r,n-r)
    result=1
    for i in range(1,r+1):
        result=result*(n-i+1)//i
    return result
print("C(5,2) =",combination(5,2))
```

```
C(5,2) = 10

...Program finished with exit code 0
Press ENTER to exit console.
```

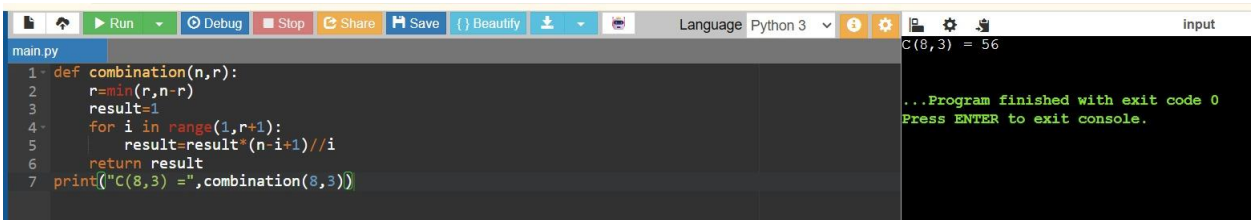
3.Write a program to find the number of ways to select r items from n items.

Input:

n = 8
r = 3

Output:

C(8,3) = 56



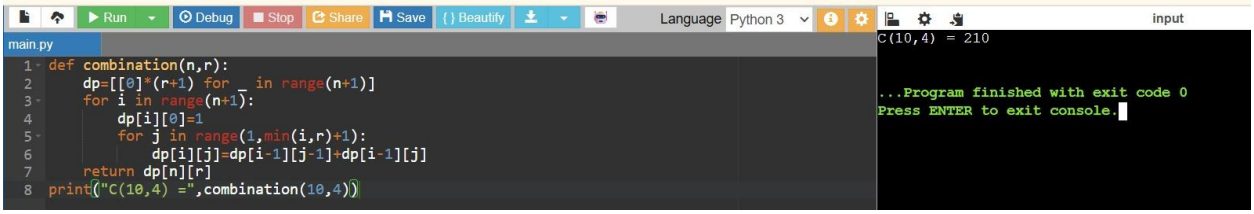
4.Write a program to compute the Binomial Coefficient using Dynamic Programming.

Input:

n = 10
r = 4

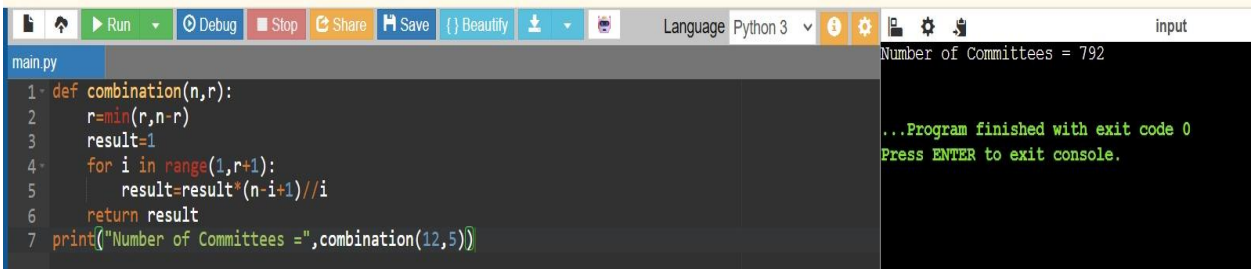
Output:

C(10,4) = 210



5.Write a program to determine the number of possible committees formed from a group.

Input:
Total Members = 12
Committee Size = 5
Output:
Number of Committees = 792



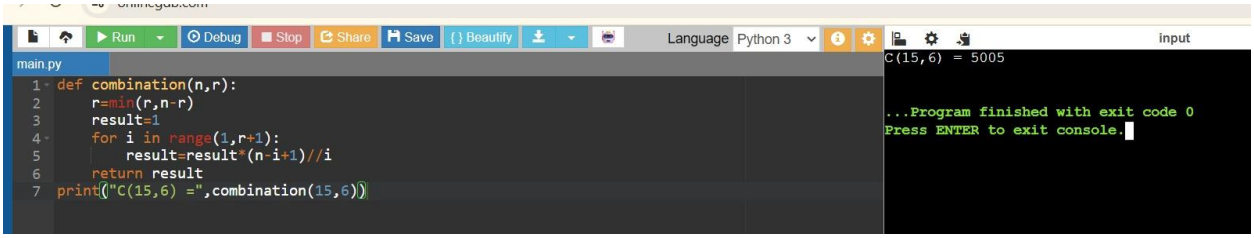
6.Write a program to calculate the combination value for given n and r.

Input:

n = 15
r = 6

Output:

C(15,6) = 5005



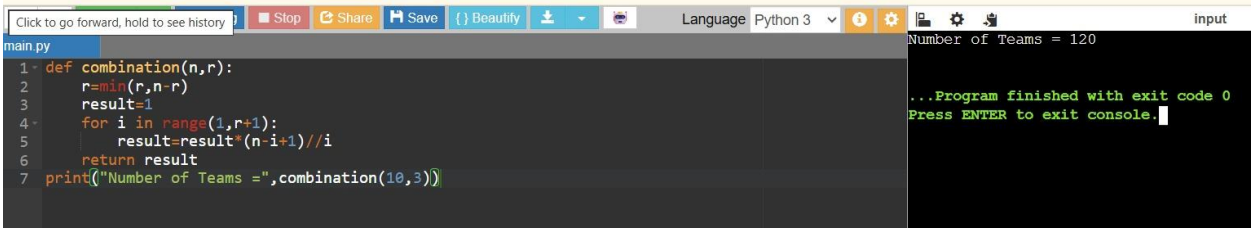
7. A software company has 10 developers and wants to select 3 developers for a special AI project. Write a program to calculate the number of different teams that can be formed.

Input:

Total Developers = 10
Developers Required = 3

Output:

Number of Teams = 120



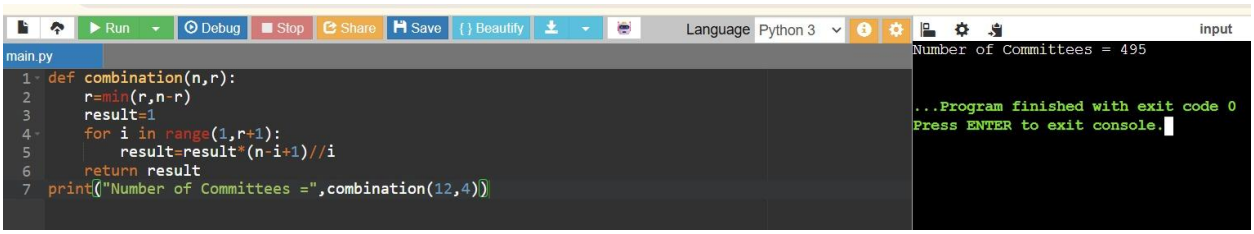
8.A university must select 4 faculty members from a pool of 12 professors to form a scholarship committee. Write a program to determine the total number of possible committees.

Input:

Total Professors = 12
Committee Size = 4

Output:

Number of Committees = 495



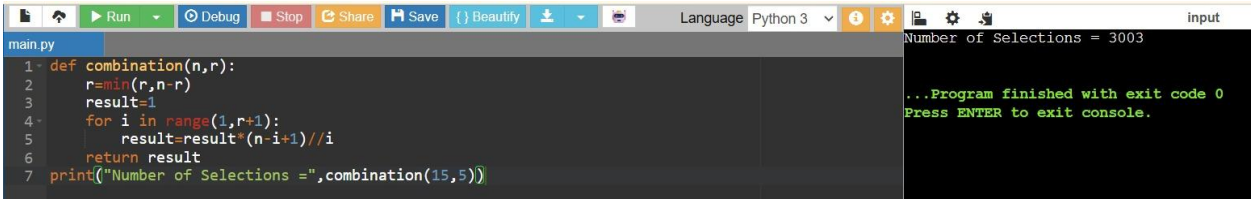
9.A cricket academy has 15 players and needs to choose 5 players for a tournament squad. Calculate the number of possible selections.

Input:

Total Players = 15
Players Required = 5

Output:

Number of Selections = 3003



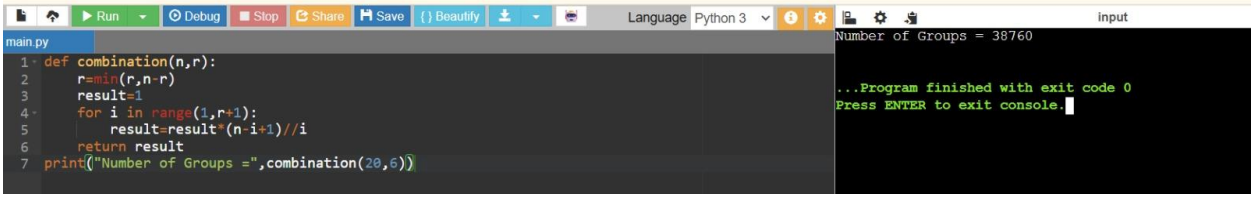
10.A company wants to select 6 customers from 20 volunteers to test a new product. Write a program to determine the total number of testing groups.

Input:

Total Volunteers = 20
Group Size = 6

Output:

Number of Groups = 38760



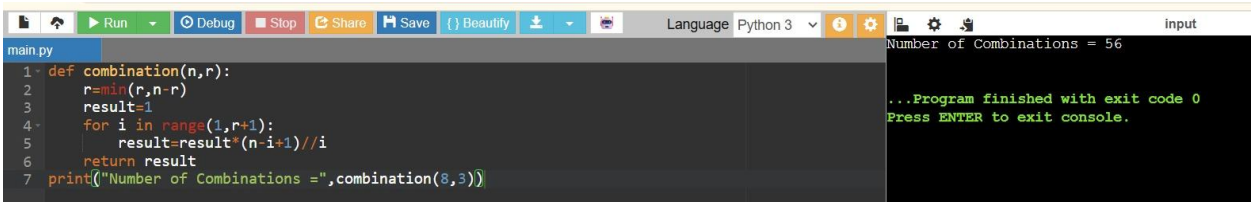
11.A cloud provider has 8 servers and must allocate 3 servers to a customer. Determine the number of possible server combinations.

Input:

Total Servers = 8
Servers Selected = 3

Output:

Number of Combinations = 56




```
def combination(n,r):
    r=min(r,n-r)
    result=1
    for i in range(1,r+1):
        result=result*(n-i+1)//i
    return result
print("Number of Combinations =",combination(8,3))
```

Assembly Line Scheduling

1.Write a program to find the minimum time required to manufacture a product using two assembly lines.

Input:

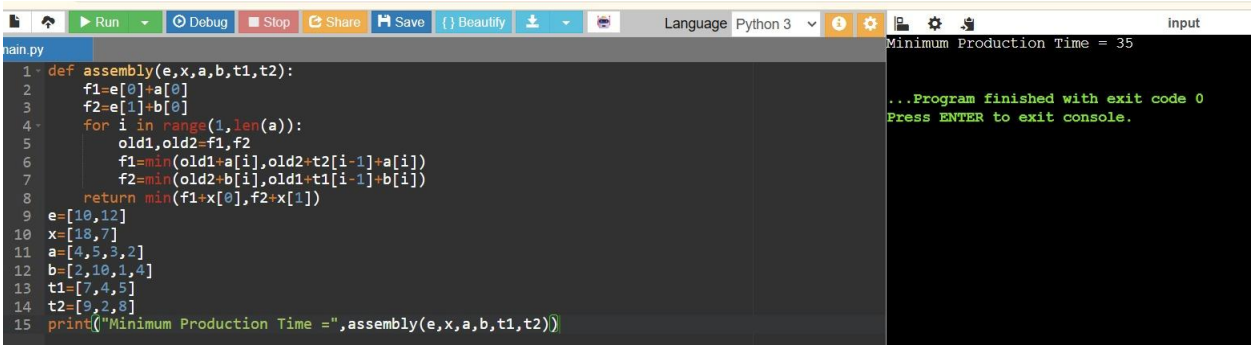
Entry Times = [10,12]
Exit Times = [18,7]

Line1 = [4,5,3,2]
Line2 = [2,10,1,4]

Transfer1 = [7,4,5]
Transfer2 = [9,2,8]

Output:

Minimum Production Time = 35




```
def assembly(e,x,a,b,t1,t2):
    f1=e[0]+a[0]
    f2=e[1]+b[0]
    for i in range(1,len(a)):
        old1,old2=f1,f2
        f1=min(old1+a[i],old2+t2[i-1]+a[i])
        f2=min(old2+b[i],old1+t1[i-1]+b[i])
    return min(f1+x[0],f2+x[1])
e=[10,12]
x=[18,7]
a=[4,5,3,2]
b=[2,10,1,4]
t1=[7,4,5]
t2=[9,2,8]
print("Minimum Production Time =",assembly(e,x,a,b,t1,t2))
```

2.Write a program to determine the optimal assembly line path with minimum processing time.

Input:

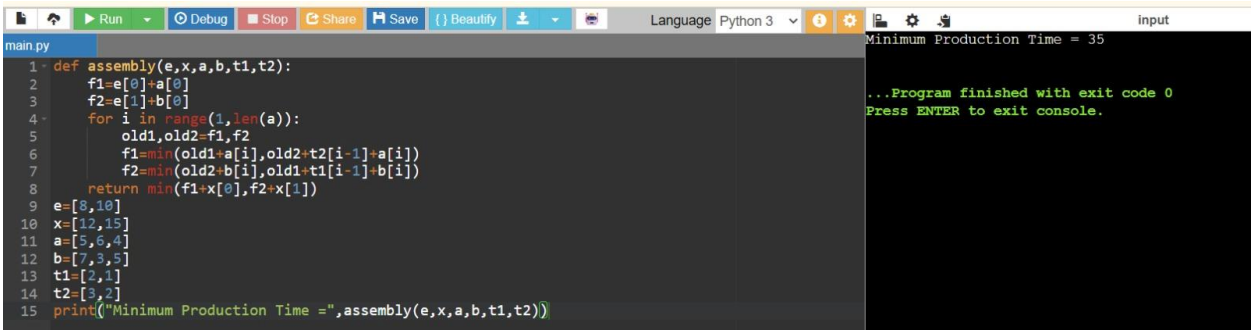
Entry Times = [8,10]
Exit Times = [12,15]

Line1 = [5,6,4]
Line2 = [7,3,5]

Transfer1 = [2,1]
Transfer2 = [3,2]

Output:

Minimum Production Time = 33



3.Write a program to compute the fastest route through assembly stations.

Input:

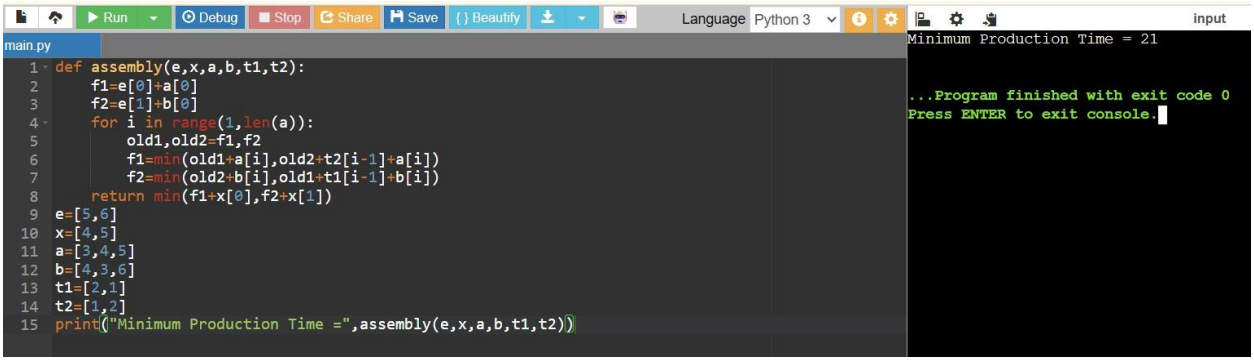
Entry Times = [5,6]
Exit Times = [4,5]

Line1 = [3,4,5]
Line2 = [4,3,6]

Transfer1 = [2,1]
Transfer2 = [1,2]

Output:

Minimum Production Time = 21



4.Write a program to identify the assembly line schedule that minimizes total manufacturing time.

Input:

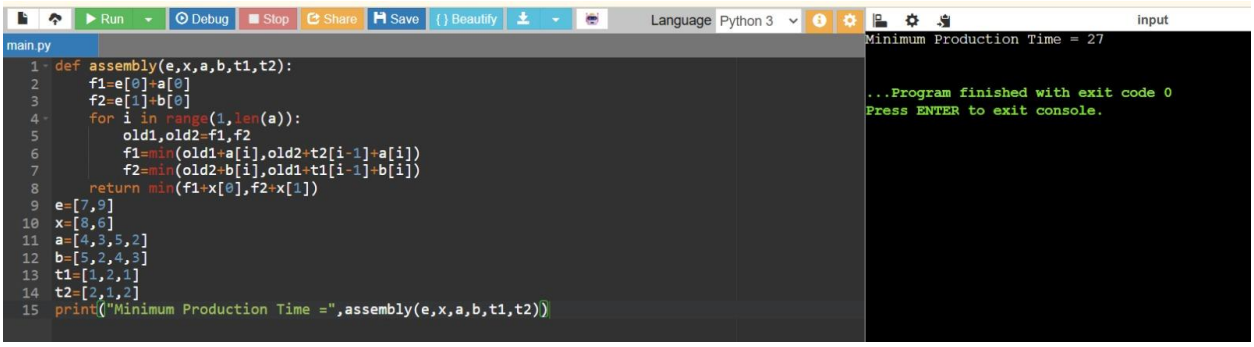
Entry Times = [7,9]
Exit Times = [8,6]

Line1 = [4,3,5,2]
Line2 = [5,2,4,3]

Transfer1 = [1,2,1]
Transfer2 = [2,1,2]

Output:

Minimum Production Time = 29



5.Write a program to determine the minimum cost path through an assembly system.

Input:

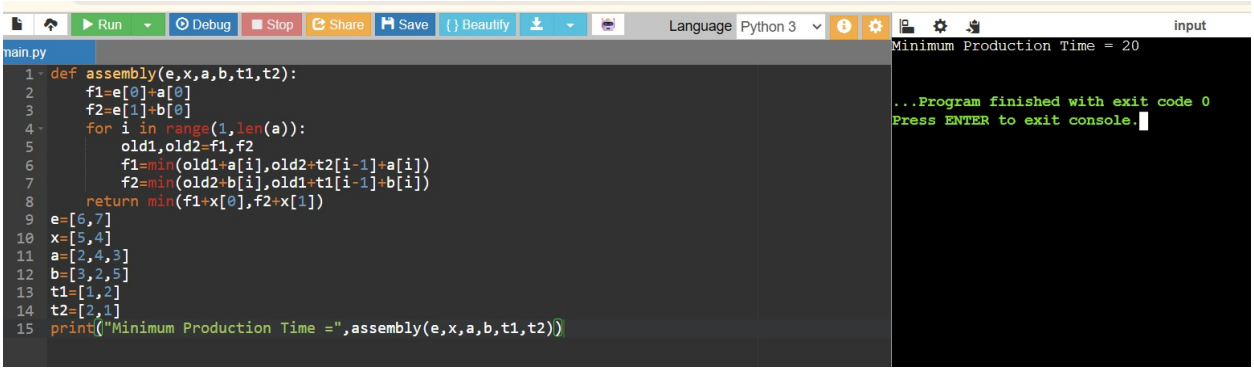
Entry Times = [6,7]
Exit Times = [5,4]

Line1 = [2,4,3]
Line2 = [3,2,5]

Transfer1 = [1,2]
Transfer2 = [2,1]

Output:

Minimum Production Time = 18



6. An automobile manufacturer has two assembly lines for producing cars. Each line contains four stations. The processing times at each station and transfer times between lines are provided. Write a program to determine the minimum time required to manufacture a car.

Input:

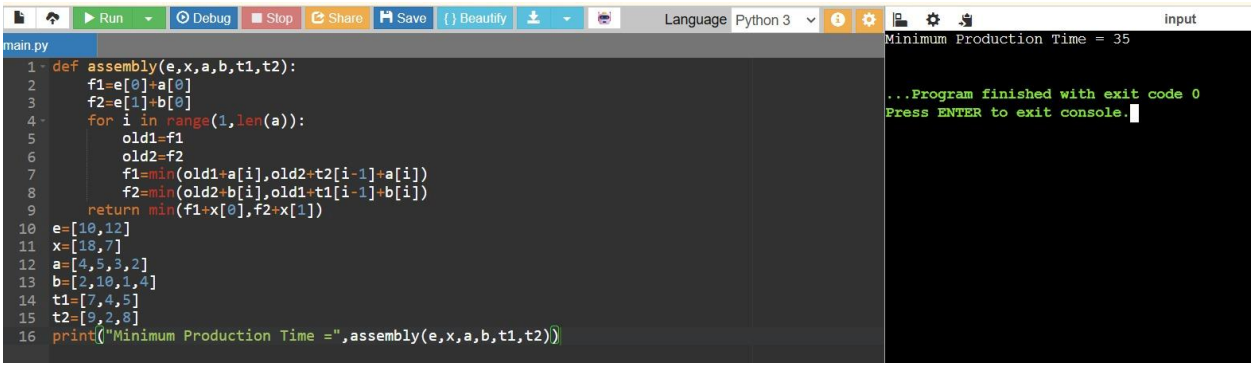
Entry Times = [10,12]
Exit Times = [18,7]

Line1 = [4,5,3,2]
Line2 = [2,10,1,4]

Transfer1 = [7,4,5]
Transfer2 = [9,2,8]

Output:

Minimum Production Time = 35



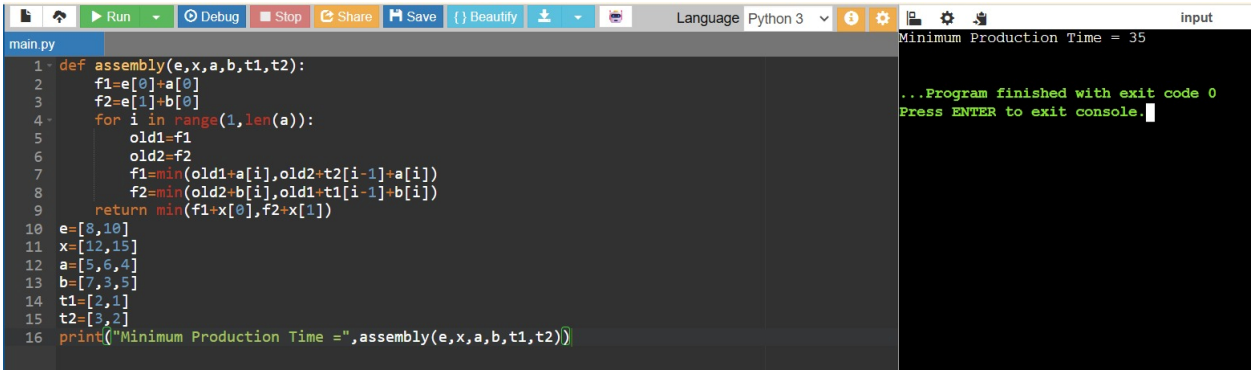
7. A smartphone company assembles devices using two production lines. Components may switch lines after completing any station. Determine the optimal route that minimizes total production time.

Input:

Entry Times = [8,10]
Exit Times = [12,15]
Line1 = [5,6,4]
Line2 = [7,3,5]
Transfer1 = [2,1]
Transfer2 = [3,2]

Output:

Minimum Production Time = 33



8.A food processing company uses two packaging lines. Products pass through labeling, sealing, and packing stations. Write a program to find the minimum completion time.

Input:

Entry Times = [5,6]
Exit Times = [4,5]

Line1 = [3,4,5]
Line2 = [4,3,6]

Transfer1 = [2,1]
Transfer2 = [1,2]

Output:

Minimum Production Time = 21

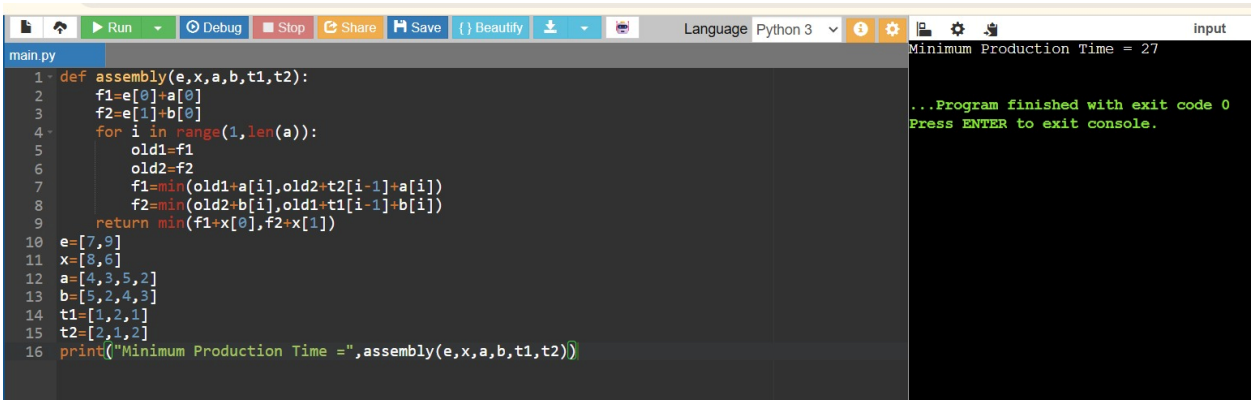
9.Medicine bottles pass through filling, capping, labeling, and packaging stations on two assembly lines. Determine the optimal production schedule.

Input:

Entry Times = [7,9]
Exit Times = [8,6]
Line1 = [4,3,5,2]
Line2 = [5,2,4,3]
Transfer1 = [1,2,1]
Transfer2 = [2,1,2]

Output:

Minimum Production Time = 27



10.A circuit board manufacturing facility uses two parallel assembly lines. Products can switch lines between stations. Find the minimum manufacturing time.

Input:

Entry Times = [6,7]
Exit Times = [5,4]
Line1 = [2,4,3]
Line2 = [3,2,5]
Transfer1 = [1,2]
Transfer2 = [2,1]

Output:
Minimum Production Time = 20